ENGINEERING

Rising High: GM Yeast Generates Known and Novel Marijuana Compounds

Engineered microorganisms churn out THC, CBD and rarer, less-understood cannabis cousins

By Sophie Bushwick on February 27, 2019

Instead of making beer, genetically-modified yeast can cook up cannabinoids. Credit: Getty Images

If you had to pick a favorite microbe, a good candidate would be *Saccharomyces cerevisiae*, better known as brewer's yeast, which transforms grape juice into wine, grain mash into beer and dough into bread. Over the past few decades scientists have hacked the yeast's genome to make it produce less delectable but arguably more important substances, including hormones like insulin and drugs like opioids. Now it is churning out cannabinoids, the compounds found in marijuana.

Researchers led by Jay Keasling, a professor of chemical engineering and bioengineering at the University of California, Berkeley, have genetically modified brewer's yeast to produce two of the most common cannabinoids, tetrahydrocannabinol (THC) and cannabidiol (CBD). They claim their method could also produce microorganisms capable of making any other naturally occurring cannabinoid as well as some brand-new varieties. Certain cannabinoids have potential as treatments for a variety of disorders, but require more research to separate hype from medical reality.

In the new study published Wednesday in *Nature*, scientists transferred known gene sequences that control metabolic pathways in cannabis plants into yeast. The resulting microorganisms can turn a sugar called galactose into intermediate chemicals, and use those chemicals to synthesize cannabigerolic acid (CBGa)—the so-called mother cannabinoid that can develop into several other compounds. Finally, each individual strain of yeast turns CBGa into a different cannabinoid. "The yeast that produces THC is different from the yeast that produces CBD, but they only differ by one gene—and that's the last gene in the pathway that takes CBGa and turns it into CBD or THC," Keasling says. "The beauty of this technology is that you can swap these out for a rare cannabinoid."

Marijuana contains more than 100 different cannabinoids, but most of them are at much lower concentrations than CBD and THC. Because plants yield very small amounts of the rarer substances, they are more expensive to produce. Even when researchers successfully extract them, the compounds are often contaminated with traces of their more common cousins. Yeast could produce purer versions of these cannabinoids, bringing the price of rare varieties to the same level as more popular ones. "It's a platform for producing all of the cannabinoids that are currently thought to exist in cannabis as well as all these unnatural ones that you'd never find in any organism," Keasling says. What makes some of these "unnatural"? Normally, marijuana plants incorporate a chemical called hexanoic acid (which humans use as a cheap food additive) into cannabinoids. When Keasling's team added different chemicals into their yeast's sugary diet, the genetically modified microbes incorporated those substances instead of the hexanoic acid, resulting in new, never-before-seen compounds.

PUTTING CANNABINOIDS TO THE TEST

"There might be a blockbuster drug or two in some of those rare ones or unnatural" cannabinoids, Keasling says. But uncovering the potential medical applications will take a lot more research—and scientists are already busy studying the effects of the more famous cannabinoids. For example, CBD shows promise for treating problems such as epilepsy, PTSD and addiction without producing the high that psychoactive THC does. The hype around CBD, however, has built up faster than the scientific research on it; proponents have made unsubstantiated claims of its ability to treat just about anything, from eczema to cancer. This promotion has gone mainstream, with coffee shops putting CBD on the menu as a supposed anxiety treatment and online vendors selling inaccurately labeled extracts.

To find the compounds' true applications, researchers must test how cannabinoids affect humans. That is a problem this new biosynthesis method may not solve, however. "It doesn't matter if CBD or any cannabinoids come from a plant or if they're made synthetically or if

they come from yeast. If the end product is still a banned substance, it doesn't increase accessibility," says Yasmin Hurd, director of the Addiction Institute at Icahn School of Medicine at Mount Sinai in New York City, who was not involved in the new study. The Drug Enforcement Administration classifies marijuana-derived substances like CBD as Schedule I drugs—a category that includes heroin. This gets in the way of researchers like Hurd, who studies how CBD impacts craving and anxiety—factors that contribute to opioid addiction. "It's really important to have more research done—it's only then that we will be able to see whether CBD or other specific cannabinoids that people are making claims about…are able to treat these symptoms and disorders," she says.

Keasling agrees that, despite looser state laws, federal restrictions on testing cannabis make research difficult—"and there's no way to get around that until [the] law changes," he says. "But what this method can do is provide some of these very rare cannabinoids that you'd never be able to extract out of cannabis, because they're produced in such small quantities. And who knows—one of those might be better than CBD or THC."

A GROWING INDUSTRY

Whether or not they help research, yeast-made cannabinoids will certainly have a commercial impact. Keasling estimates his method could produce cannabinoids at a cost equal to or lower than that required by agricultural marijuana production. He founded the company Demetrix to license his new technology from Berkeley and develop commercial cannabinoid production. And he has plenty of competition. Last year biotechnology start-ups Librede and Gingko Bioworks announced, respectively, a patent for producing CBD from yeast and a multimillion-dollar partnership with cannabis company Cronos Group. Demetrix CEO Jeff Ubersax estimates that about 15 to 20 other companies are competing to turn yeast cells into little cannabinoid factories.

Yet Keasling claims his team is the first to develop a process that relies only on sugar—his yeast can make cannabinoids from galactose alone without requiring additional, more expensive ingredients. Furthermore, Ubersax says the new yeast produces cannabinoids at a rate several orders of magnitude higher than that of other strains. "Prior work had identified the basic parts you need but was producing them at a very low level," he says, comparing the output with that of a compact car engine. Keasling, he says, "identified a new DNA segment that's more like a jet engine."

Still, this explosion of start-ups will not help investigators like Hurd any more than the rise of CBD lattes has. "The problem is that companies are not putting any money investments into research," she says. They are not contributing anything to the advancement of

knowledge, she adds. Without further study the federal government will not sanction more uses for cannabinoids—the U.S. Food and Drug Administration has only approved one drug containing substances derived from marijuana (the epilepsy medication Epidiolex, which includes CBD). Hurd has an idea to break this cycle. “Tax the companies and put that tax money toward research,” she says. “For companies that want to make a lot of money in CBD, say, ‘Okay, this is your contribution.’ And let’s just get the research done so [the federal government] can approve more.”

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